Containerization-and-DevOps

🐳 Class Task

🛠️ Tools Used

- Docker
- k3d
- kubectl
- WSL

⚙️ Steps

Download kubectl (Initial Attempt)

```
curl -LO https://dl.k8s.io/release/stable/bin/linux/amd64/kubectl
```

Fix Download Using Correct Version

```
VERSION=$(curl -L -s https://dl.k8s.io/release/stable.txt)

curl -LO https://dl.k8s.io/release/$VERSION/bin/linux/amd64/kubectl
```

Make kubectl Executable

```
chmod +x kubectl
sudo mv kubectl /usr/local/bin/
```

Verify kubectl Installation

```
kubectl version --client
```

Install k3d

```
curl -s https://raw.githubusercontent.com/k3d-io/k3d/main/install.sh | bash
```

Verify k3d Installation

```
k3d version
```

Create Kubernetes Cluster

```
k3d cluster create mycluster
```

Check Cluster Nodes

```
kubectl get nodes
```

Check Context

```
kubectl config get-contexts
```

⚠️ Notes

- Initial kubectl download failed due to incorrect URL
- Correct version fetched dynamically using `stable.txt`
- k3d creates lightweight Kubernetes cluster using Docker
- kubectl is used to interact with the cluster

✅ Outcome

- Installed kubectl successfully
- Installed k3d and created cluster
- Verified cluster node status
- Confirmed Kubernetes context configuration

📸 Screenshots

```
sristee@Sristee:/mnt/c/Users/srist$ curl -LO https://dl.k8s.io/release/stable/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   245  100   245    0     0    272      0 --:--:-- --:--:-- --:--:--   272
sristee@Sristee:/mnt/c/Users/srist$ chmod +x kubectl
sristee@Sristee:/mnt/c/Users/srist$ sudo mv kubectl /usr/local/bin/
sristee@Sristee:/mnt/c/Users/srist$ kubectl version --client
/usr/local/bin/kubectl: line 1: syntax error near unexpected token `<'
/usr/local/bin/kubectl: line 1: `<?xml version='1.0' encoding='UTF-8'?><Error><Code>NoSuchKey</Code><Message>The specified key does n
ot exist.</Message><Details>No such object: 767373bbdcb8270361b96548387bf2a9ad0d48758c35/release/stable/bin/linux/amd64/kubectl</Deta
ils></Error>'
sristee@Sristee:/mnt/c/Users/srist$ curl -s https://raw.githubusercontent.com/k3d-io/k3d/main/install.sh | bash
Preparing to install k3d into /usr/local/bin
k3d installed into /usr/local/bin/k3d
Run 'k3d --help' to see what you can do with it.
sristee@Sristee:/mnt/c/Users/srist$ sudo rm /usr/local/bin/kubectl
sristee@Sristee:/mnt/c/Users/srist$ rm kubectl
rm: cannot remove 'kubectl': No such file or directory
sristee@Sristee:/mnt/c/Users/srist$ VERSION=$(curl -L -s https://dl.k8s.io/release/stable.txt)
sristee@Sristee:/mnt/c/Users/srist$ curl -LO https://dl.k8s.io/release/$VERSION/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 55.8M  100 55.8M    0     0  1380k      0  0:00:41  0:00:41 --:--:-- 1368k
sristee@Sristee:/mnt/c/Users/srist$ chmod +x kubectl
sudo mv kubectl /usr/local/bin/
sristee@Sristee:/mnt/c/Users/srist$ kubectl version --client
Client Version: v1.35.2
Kustomize Version: v5.7.1
sristee@Sristee:/mnt/c/Users/srist$ k3d version
k3d version v5.8.3
k3s version v1.31.5-k3s1 (default)
```

```
sristee@Sristee:/mnt/c/Users/srist$ k3d cluster create mycluster
INFO[0000] Prep: Network
INFO[0000] Created network 'k3d-mycluster'
INFO[0000] Created image volume k3d-mycluster-images
INFO[0000] Starting new tools node...
INFO[0001] Creating node 'k3d-mycluster-server-0'
INFO[0004] Pulling image 'ghcr.io/k3d-io/k3d-tools:5.8.3'
INFO[0005] Pulling image 'docker.io/rancher/k3s:v1.31.5-k3s1'
INFO[0018] Starting node 'k3d-mycluster-tools'
INFO[0061] Creating LoadBalancer 'k3d-mycluster-serverlb'
INFO[0064] Pulling image 'ghcr.io/k3d-io/k3d-proxy:5.8.3'
INFO[0085] Using the k3d-tools node to gather environment information
INFO[0086] HostIP: using network gateway 172.20.0.1 address
INFO[0086] Starting cluster 'mycluster'
INFO[0086] Starting servers...
INFO[0086] Starting node 'k3d-mycluster-server-0'
INFO[0090] All agents already running.
INFO[0090] Starting helpers...
INFO[0090] Starting node 'k3d-mycluster-serverlb'
INFO[0096] Injecting records for hostAliases (incl. host.k3d.internal) and for 2 network members into CoreDNS configmap...
INFO[0098] Cluster 'mycluster' created successfully!
INFO[0098] You can now use it like this:
kubectl cluster-info
sristee@Sristee:/mnt/c/Users/srist$ kubectl get nodes
NAME                     STATUS   ROLES                  AGE   VERSION
k3d-mycluster-server-0   Ready    control-plane,master   22s   v1.31.5+k3s1
sristee@Sristee:/mnt/c/Users/srist$ kubectl config get-contexts
CURRENT   NAME            CLUSTER         AUTHINFO              NAMESPACE
*         k3d-mycluster   k3d-mycluster   admin@k3d-mycluster
sristee@Sristee:/mnt/c/Users/srist$
```